

BUSINESS PROBLEM STATEMENT

Consumer Shopping Behavior Analytics Project — Retail Sector

Business Context

A leading retail company wants to deepen its understanding of customer shopping behavior in order to strengthen sales performance, elevate customer satisfaction, and build long-term loyalty. The initiative is grounded in the following observations and priorities:

- **Shifting purchasing patterns:** Management has observed noticeable shifts in how customers buy — across demographic groups (age, gender, location, income level), across product categories, and across sales channels, spanning both online and offline (in-store) touchpoints.
- **Cross-channel complexity:** The rise of parallel online and offline buying journeys makes it harder to see a unified view of the customer, making it difficult to know where to focus marketing spend and inventory investment.
- **Unclear purchase drivers:** It is not yet clear which factors most influence a customer's decision to buy and to come back — candidates include promotional discounts, product reviews and ratings, seasonal effects, and preferred payment methods.
- **Loyalty and retention risk:** Without a clear read on what drives repeat purchases, the company risks losing customers to competitors who are better able to personalize offers and anticipate demand.
- **Need for a data-driven narrative:** Leadership wants these behavioral signals translated into a coherent, evidence-based narrative that can directly inform marketing, merchandising, and customer-retention strategy.

Overarching Business Question

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Key Objectives

- **Identify trends** — Detect meaningful patterns in shopping behavior across demographics, categories, and channels over time.
- **Improve customer engagement** — Understand what keeps customers coming back so engagement efforts can be targeted rather than generic.
- **Optimize marketing strategy** — Direct promotional spend toward the discounts, channels, and moments that demonstrably move purchase behavior.
- **Strengthen product strategy** — Use category- and season-level insight to guide assortment, pricing, and inventory decisions.

Project Deliverables

- **1. Data Preparation & Modeling (Python):** Ingest the raw consumer shopping dataset and perform end-to-end cleaning: handle missing values, remove duplicates, correct data types, and standardize categorical fields (e.g., channel, payment method, category). Engineer derived features such as customer recency/frequency/monetary (RFM) metrics, discount-usage flags, and seasonal indicators to prepare an analysis-ready dataset for downstream modeling.
- **2. Data Analysis (SQL):** Design a structured relational schema (customers, transactions, products, channels) and load the cleaned data to simulate real business transactions. Write queries to segment customers, measure repeat-purchase and loyalty rates, and quantify the impact of discounts, reviews, seasonality, and payment preferences on purchase behavior.
- **3. Visualization & Insights (Power BI):** Build an interactive, stakeholder-facing dashboard that surfaces key trends — sales by channel and category, customer segments, discount effectiveness, and seasonal demand patterns — with drill-down filters that support fast, data-driven decision-making.
- **4. Report and Presentation:** Produce a clear, well-structured project report summarizing methodology, key findings, and actionable business recommendations, supported by a concise stakeholder presentation that visually communicates the insights and their business impact.
- **5. GitHub Repository:** Consolidate all Python notebooks/scripts, SQL schema and queries, Power BI files, and supporting documentation into a well-organized, version-controlled repository with a clear README for reproducibility.

Goal: Convert raw consumer shopping data into trend insights, stronger customer engagement, and optimized marketing and product strategy.